

Team Number :	apmcm2213912
Problem Chosen :	E

ARIMA-based prediction of the number of future nuclear weapons and its implications for the human survival environment

Abstract

As we all know the explosion of an atomic bomb could destroy a city. Because of the terrible power of the bomb, many countries want to use it as a deterrent to other countries. While nuclear bombs serve as a political deterrent due to their terrifying power, they also raise concerns that nuclear weapons could destroy the earth due to its power. In this paper, we analyze the past and future of the number of nuclear weapons by processing the data with Stata and Python.

For question 1, by analyzing the four tables of the underlying data, we found that there were ten countries that once possessed nuclear weapons. In the last 20 years, the country with the largest decrease in nuclear weapons is the United States, while the country with the largest increase in nuclear weapons is Pakistan. From 1962-1966, the number of nuclear tests in the world reached 422. North Korea who has been the most active country in nuclear research in the last decade. The United States has taken the least time to move from "not considering possession and weapons" to "possessing nuclear weapons".

For question 2, firstly we made assumptions about the future world situation and future development trends, and our assumptions are relatively optimistic. Countries will use their nuclear weapons with great caution, and *Treaty on the Non —Proliferation of Nuclear Weapons* will play a positive role continuously. Then we make predictions through multiple models and come up with an R^2 of 0.975 for the ARIMA model, which is higher than the LSTM and CEEMDAN-LSTM. Finally, we discovered that there are 9 nuclear-armed countries in the future. Globally, the trend of the total number of nuclear weapons in the future is decreasing, and the trend of the number of nuclear weapons in each country is consistent with the global.

For question 3, We have defined the destruction of the Earth by nuclear bombs. When nuclear weapons can destroy 1% of human habitable land area, it means destroying the human living environment, and when destroying 5%, it means completely destroying the earth. Subsequently, after calculation, we found that at least 5241 nuclear bombs are needed to completely destroy the human living environment. While the existing nuclear bombs can destroy 9% of the habitable land area, which is sufficient to destroy the human living environment. For the safety of human survival, we gave a nuclear weapons limit of 1048 bombs in total, and each nuclear-armed country was assigned a limit of 11% of the existing number of nuclear weapons.

For question 4, based on the above models and calculations, we wrote a non-technical paper on our findings and submitted it to the United Nations with recommendations for each nuclear-armed country.

Key Words: ARIMA, LSTM, CEEMDAN-LSTM, nuclear weapons, human survival

Contents

1 Questions Restatement	2
1.1 Questions Background.....	2
1.2 Questions to ask.....	2
1.2.1 Question 1	2
1.2.2 Question 2	3
1.2.3 Question 3	3
1.2.4 Question 4	3
2 Questions Analysis	4
3 Model Assumptions and Notation.....	5
3.1 Model Assumptions	5
3.2 Symbol Description	5
4 Modeling and problem solving	6
4.1 Solution of Questions 1	6
4.1.1 Sub-question 1	6
4.1.2 Sub-question 2	6
4.1.3 Sub-question 3	7
4.1.4 Sub-question 4	8
4.1.5 Sub-question 5	9
4.2 Solution of Questions 2	10
4.2.1 Sub-question 1	10
4.2.2 Sub-question 2	16
4.3 Solution of Questions 3	19
4.3.1 Sub-question 1	19
4.3.2 Sub-question 2	20
4.3.3 Sub-question 3	20
4.4 Solution of Questions 4	22
References.....	23
Appendix.....	24

1 Questions Restatement

1.1 Questions Background

On August 6, 1945, near the end of World War II, the United States dropped an atomic bomb called "Little Boy" on Hiroshima, Japan, killing 200,000 people. The explosion yield of an atomic bomb is about tens to hundreds of thousands of tons of TNT equivalent. The explosion of an atomic bomb and its radiation range can destroy a city.

Because of the terrible power of the atomic bomb, many countries wanted to use it to deter other countries and protect their own countries from foreign invasion. Nuclear deterrence is the declared or implied use of nuclear weapons to create terror and psychological pressure to make the other side change their intentions or stop their actions^[1]. The existence of nuclear weapons has given some countries the ability to deter others from using nuclear weapons to blackmail them and to defend themselves against nuclear aggression^[2].

Nuclear bombs serve as a political deterrent because of their frightening power, but also because of their great power, there is concern that nuclear weapons could destroy the Earth. Therefore, estimating the number of nuclear bombs needed to destroy the Earth and predicting future global trends in nuclear weapons has become an important task for nuclear security personnel.

1.2 Questions to ask

A collection of global nuclear weapons-related data is available, and the paper is asked to address the following related questions:

1.2.1 Question 1

Required to analyze the underlying data and answers to the following questions.

(1) Which country's nuclear weapons stockpile has decreased or increased the most in the past 20 years?

- (2) In which five years did the most nuclear weapons testing occur?
- (3) Which country has been most active in nuclear weapons research in the past 10 years?
- (4) Which country has made the fastest transition from "not considering nuclear weapons" to "possessing nuclear weapons"?

1.2.2 Question 2

Required to predict the number of nuclear weapons to meeting the following requirements.

- (1) Develop a mathematical model to predict the number of nuclear weapons and predict the countries that will have nuclear weapons in the next 100 years based on the data given or information collected on their own.
- (2) Predict the trend in the number of nuclear weapons over the next 100 years, the total number of nuclear weapons in 2123, and the number of nuclear weapons in each country.

1.2.3 Question 3

Required to protect our planet, and answer the following questions.

- (1) Create a mathematical model of the detonation location of a nuclear weapon and calculate the minimum number of nuclear bombs needed to destroy the Earth?
- (2) Based on the mathematical model, what is the maximum destructive power that a nuclear bomb currently possesses? Is it enough to destroy the Earth?
- (3) To protect the Earth and the environment we depend on, what should the total number of nuclear bombs in the world be limited to, and what should be the theoretical limit on the number of countries that already have nuclear weapons?

1.2.4 Question 4

It is required to prepare a non-technical article explaining your team's findings to the United Nations and making several recommendations for all countries.

2 Questions Analysis

Question 1 requires an analysis of the base data to answer five sub-questions. **For sub-question 1**, We should find the country with "had possessed" status from the underlying data; **For sub-question 2**, we should count the changes between the nuclear weapons stockpile of each country in 2022 and 2002; **For sub-question 3**, we should add up the number of nuclear weapons worldwide for each year and the following four years, and then find which one has the most nuclear weapons. **For sub-question 4**, the number of nuclear weapons experiments for each country from 2012 to 2022 should be added together to analyze how active each country; **For sub-question 5**, you should calculate the number of years it took for each country to go from "not considering nuclear weapons" to "having nuclear weapons" and conduct a study on which country has made the fastest transition.

Question 2 asks for a prediction of the number of nuclear weapons and answers two sub-questions. **For sub-question 1**, the first step is to predict the future number of nuclear weapons for the nuclear-armed states through a time series assuming future global political developments, and for the non-nuclear-capable states through other factors to predict whether they will possess nuclear weapons. **For sub-question 2**, the model of sub-question 1 can predict the trend of nuclear weapons in the next 100 years. And the total number of nuclear weapons in 2123 is forecasted, as well as the number of nuclear weapons in each country.

Question 3 asks to protect our planet by answering three sub-questions. **For sub-question 1**, you should first assume the definition of "nuclear bombs destroy the Earth", then quantify the damage of each nuclear bomb to the Earth, and then estimate how many nuclear bombs can destroy the Earth; **For sub-question 2**, you can calculate the damage to the Earth based on the number of existing nuclear bombs according to the model in sub-question 1, and determine whether it is enough to destroy the Earth. **For sub-question 3**, firstly we should assume the Earth's resources necessary for human survival, and then calculate how much more harm nuclear weapons can do, after that we should calculate a limit on the total number of nuclear bombs in the world, finally

assign this limit based on the current ratio of nuclear bombs in each country.

Question 4 requires to prepare of a non-technical paper presenting our findings to the United Nations, as well as making recommendations.

3 Model Assumptions and Notation

3.1 Model Assumptions

For the sake of scientific and rigorous modeling, we made the following assumptions.

- (1) **Assumption 1:** The underlying data given in the question is completely correct, there are no error terms, and it is ready to be used directly without data cleaning.
- (2) **Assumption 2:** Assuming a more moderate political situation in the future and the continued role of the Nuclear Non-Proliferation Agreement, which prohibits the number of nuclear tests by member states, having a nuclear capability is very difficult.
- (3) **Assumption 3:** The meaning of nuclear destruction of the Earth is that nuclear bombs destroy the environment on which human beings depend, specifically all land areas on the Earth that are suitable for human existence, and the area of harm of each nuclear bomb is the area of land it can completely destroy.
- (4) **Assumption 4:** Considering the population distribution on Earth, if a nuclear bomb can destroy most of the places where humans live, then it is called destroying the Earth, which is about 5% of the habitable area of human beings.
- (5) **Assumption 5:** In order to protect the human environment, the total number of nuclear bombs should be guaranteed to cause devastating damage to at most 1% of the human habitable area.

3.2 Symbol Description

The symbolic descriptions in this paper are listed as shown in Table 1.

Table 1: Symbol Description

Symbols	Symbol Meaning	Unit
Country	Country Name	-
Abbreviation	Country Abbreviation	-
First time	Year of first nuclear ownership	Year

Number of changes	Number of nuclear weapons changes	pc
Test	Number of nuclear tests	time
Time	Year of change	Year
AIC	Akachi Information	-
BIC	Bayesian Information	-
Df Residuals	Degree of freedom	1
N	Number of observations	1
R ²	Goodness of fit	%
Stockpiles	Nuclear weapons possession	pc
Livable Proportion	Proportion of human habitable area	%
Livable Area	Human habitable area	10,000 km2
Area destroyed	Area of destruction	10,000 km2
Available Quantity	Nuclear possession in 2022	pc
Limit the number of	Theoretical Limit on Nuclear Holdings	pc

4 Modeling and problem solving

4.1 Solution of Questions 1

4.1.1 Sub-question 1

By analyzing the position table, we counted the countries that have ever possessed nuclear weapons, and the results are shown in Table 2. There are 10 countries in total, in chronological order, United States, Russia, United Kingdom, France, China, Israel, South Africa, India, Pakistan, and North Korea.

Table 2: Countries possessed nuclear weapons

Country	Abbreviation	First time
United States	USA	1945
Russia	RUS	1949
United Kingdom	GBR	1952
France	FRA	1960
China	CHN	1964
Israel	ISR	1967
South Africa	ZAF	1979
India	IND	1987
Pakistan	PAK	1987
North Korea	PRK	2006

4.1.2 Sub-question 2

By analyzing the stockpiles table, the number of nuclear weapons possessed by

each country in 2022 was subtracted from the number in 2002 to obtain the number of nuclear weapons changes in each nuclear-armed country in the last 20 years, and the results are shown in Table 3 and Figure 1. It can be seen that the country with the largest decrease in the number of nuclear weapons is the United States, with a decrease of 6749, a decrease of 64.54% compared to 2002, while the country with the largest increase in the number of nuclear weapons is Pakistan, with an increase of 139, an increase of 534.62% compared to 2002.

Table 3: The number of nuclear weapon changes

Country	Abbreviation	Number of changes
United States	USA	-6749
Russia	RUS	-5637
United Kingdom	GBR	-100
France	FRA	-60
South Africa	ZAF	0
Israel	ISR	14
North Korea	PRK	20
China	CHN	115
India	IND	137
Pakistan	PAK	139

4.1.3 Sub-question 3

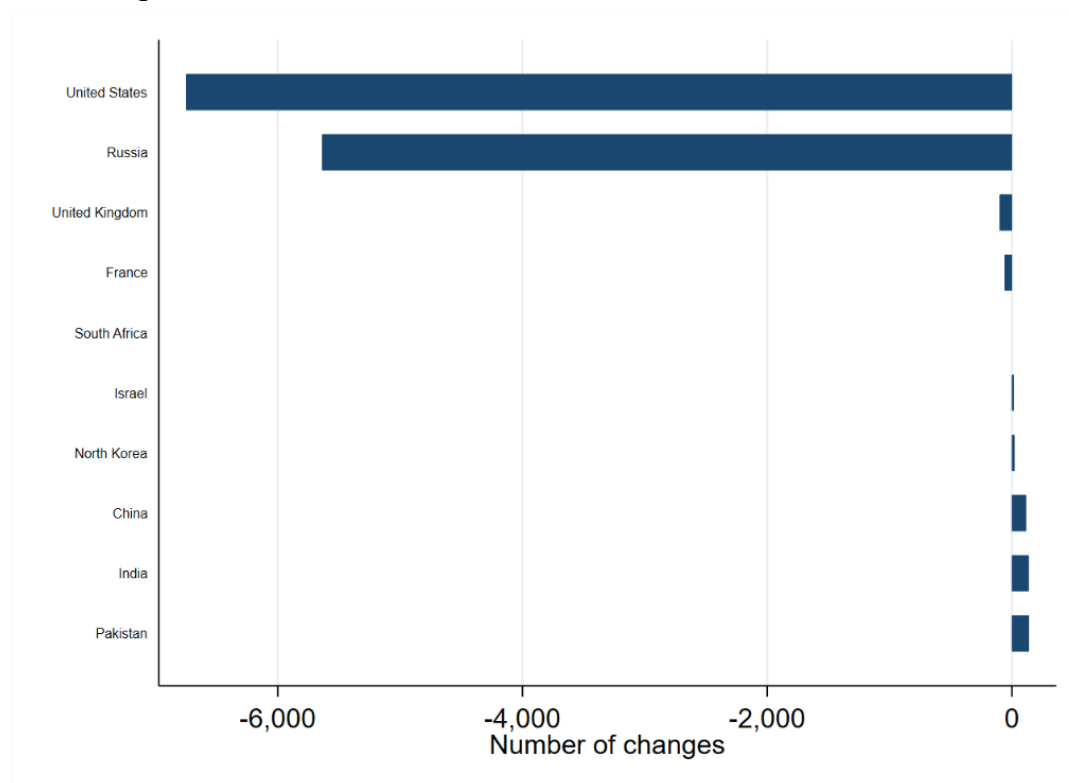


Figure 1: The number of nuclear weapon changes

We calculated the total number of nuclear tests worldwide for each year from 1945 to 2015 through a table, and then used this data to calculate the total number of nuclear tests every five years., and plotted Figure 2. The specific figures are shown in Table 4.

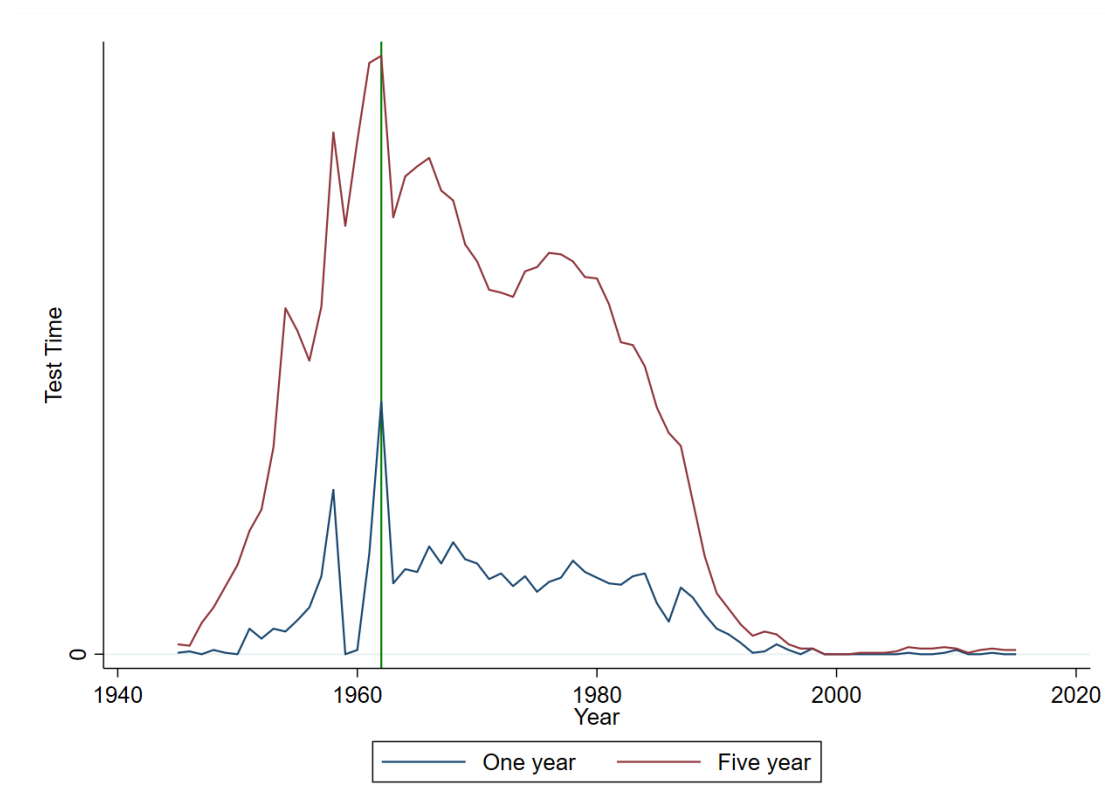


Figure 2: The total number of nuclear tests per five years

Table 4: The total number of nuclear tests per five years

Year	Test	Sum
1962	178	178
1963	50	228
1964	60	288
1965	58	346
1966	76	422

4.1.4 Sub-question 4

We summed up the number of nuclear tests in the tests table from 2012-2022 by each country separately to get the total number of nuclear tests in each country in the last decade, and the results are shown in Table 5. We can find that only North Korea conducted four nuclear tests in this last decade, and none of the other nuclear-armed countries or nuclear threshold countries conducted nuclear tests, so we consider North Korea to be the most active country in nuclear research in the last decade.

Table 5: Total number of nuclear tests by nuclear-armed countries last decade

Country	Abbreviation	Test
India	IND	0
France	FRA	0
Russia	RUS	0
United Kingdom	GBR	0
United States	USA	0
Pakistan	PAK	0
China	CHN	0
North Korea	PRK	4

4.1.5 Sub-question 5

According to the question, we subtracted the earliest year of possessing nuclear weapons from the latest year of not considering nuclear weapons for each country in the position table, which means we got the time it took for each country to change from not considering possessing nuclear weapons to possessing nuclear weapons, and the result is shown in Table 6 and Figure 3, we can find that the fastest change happened in the United States, which only took 7 years.

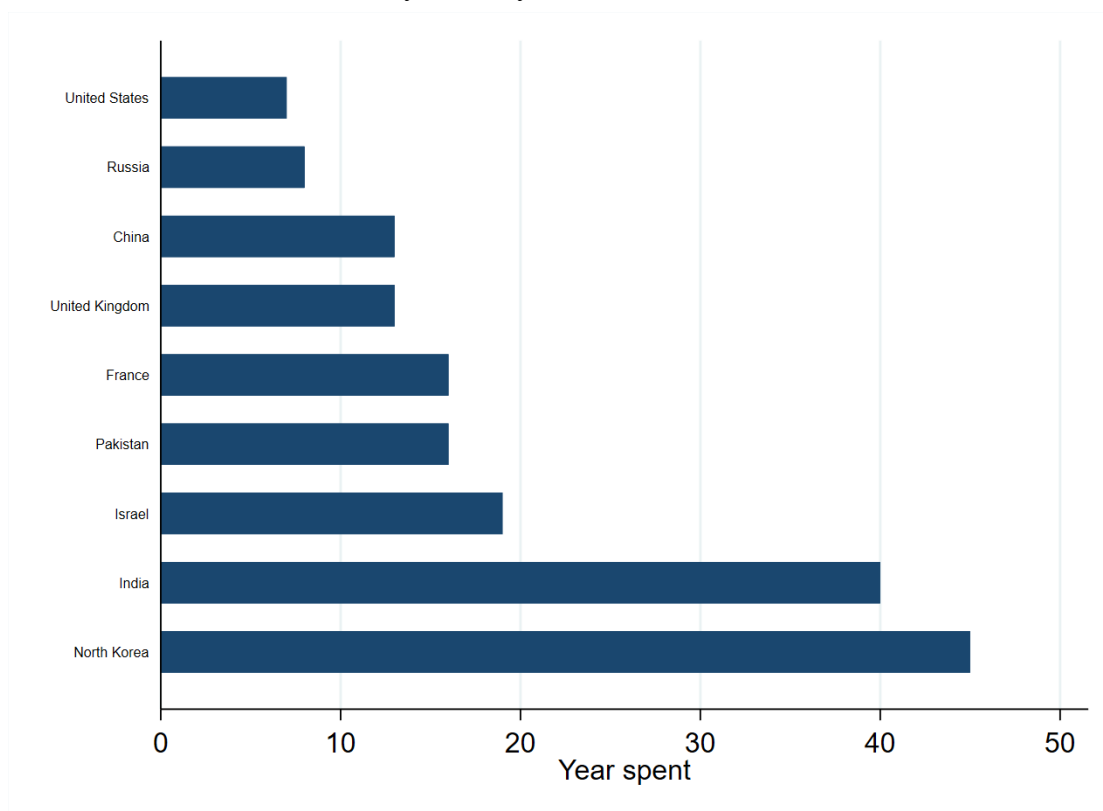
*Figure 3: Time from “disregard” to “nuclear congestion”*

Table 6: Time for a change in attitude towards nuclear weapons

Country	Does not consider	Considers	Time
United States	1938	1945	7
Russia	1941	1949	8
United Kingdom	1939	1952	13
China	1951	1964	13
Pakistan	1971	1987	16
France	1944	1960	16
Israel	1948	1967	19
India	1947	1987	40
North Korea	1961	2006	45

4.2 Solution of Questions 2

4.2.1 Sub-question 1

Before addressing this issue, please allow us to set out our thinking on this matter.

First of all, it is well known that on July 1, 1968, an international treaty, *Treaty on the Non-Proliferation of Nuclear Weapons* (NPT), was signed in London, Washington and Moscow. The purpose of the treaty is to prevent nuclear proliferation, promote nuclear disarmament and foster international cooperation in the peaceful uses of nuclear energy. The treaty entered into force in March 1970. It can be found that the United States and Russia (the former Soviet Union) both reduced their nuclear weapons by a considerable amount after the Treaty came into force, which of course has something to do with the end of the Cold War, at the same time, looking at the situation of all countries in the world, it can be said that the Treaty has effectively stopped the misuse of nuclear weapons by human beings, and the countries possessing nuclear weapons are controlled to a certain extent, which is beneficial to the future development of human being. Therefore, we have reason to believe that the treaty will continue to play a restraining role in the development of nuclear weapons in the future.

Second, the trend of the world is peace and development, with the further development of globalization which is irreversible, human beings have been very cautious about the political decision to wage war, let alone the use of nuclear weapons, and in the recent outbreak of the still ongoing war between Russia and Ukraine, Russia complied with the treaty and did not use nuclear weapons against Ukraine, so we can

believe that as history moves forward, nuclear weapons will be more of a deterrent. The future trend of whole world should to be reduce the number of nuclear weapons, while non-nuclear-capable countries will continue to have difficulty in possessing nuclear weapons to become nuclear-armed countries for a long time in the future.

Finally, considering the availability of data and the applicability of the models, we finally decided to use ARIMA, LSTM and CEEMDAN-LSTM to forecast the future number of nuclear weapons, and the main idea is to use the data of the first 60 years as the training set and the data of the last 12 years as the test set, and to compare the correctness of the fit of each model to decide which model is finally used for the forecast.

(1) ARIAM model

The ARIMA model is a forecasting method which is means widely used in time series forecasting and has gained popularity for its simplicity and stability^[3]. The model is a mathematical model for trend extrapolation by statistical analysis using the study of time series information and following the cyclical pattern of changes in the occurrence of historical events^[4]. The time series analysis method is also known as time series forecasting method, historical extrapolation method or extrapolation method. Its basic expression is ARIMA (P, D, Q), where P is the number of autoregressive terms, Q is the number of sliding average terms, and D is the number of differences (orders) made to make it a smooth series^[5].

To use the ARIMA model, first, we need to ensure that the series being predicted is smooth, therefore, we need to perform a unit root test, and the results are shown in Table 7. It can be found that the original series and the first-order difference series do not satisfy the assumption of serial smoothness at 10% confidence level, while the series after the second-order difference satisfies the assumption of smoothness at 1% confidence level, so the parameter D in ARIMA is set to 2.

Table 7: Robustness test of difference series

Projects	T-value	1%	5%	10%
Original sequence	-1.172	-3.559	-2.918	-2.594
First order differential	-1.952	-3.560	-2.919	-2.594
Second order differential	-7.340***	-3.562	-2.920	-2.595

Next, we need to determine the parameter P and Q values and the model to be applied based on the autocorrelation and partial autocorrelation plots. Figures 4 and 5 show the autocorrelation and partial autocorrelation plots of the sequence after second-order differencing, respectively.

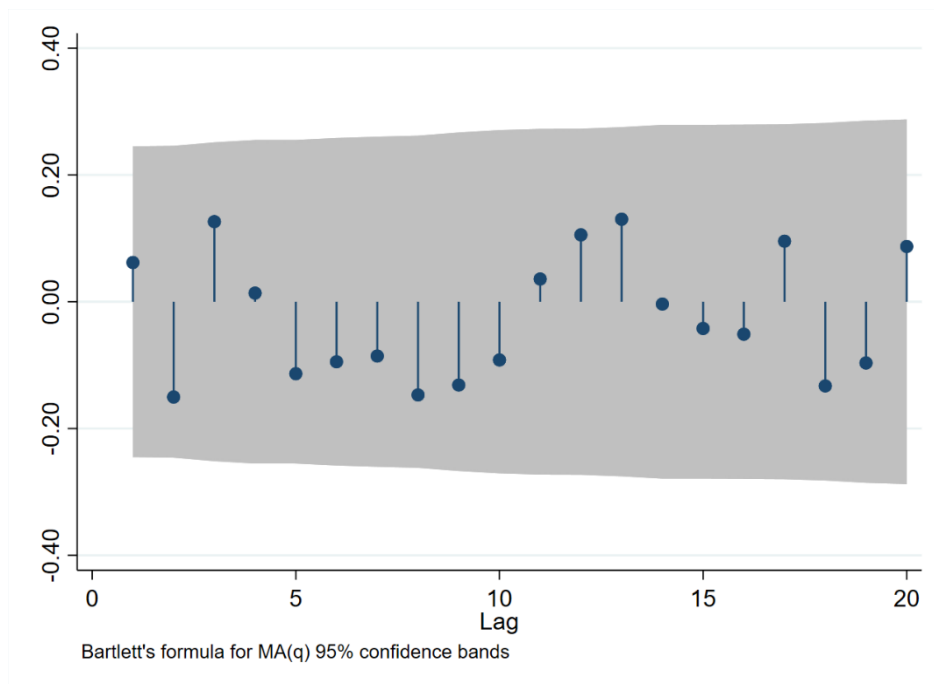


Figure 4: Autocorrelation Chart

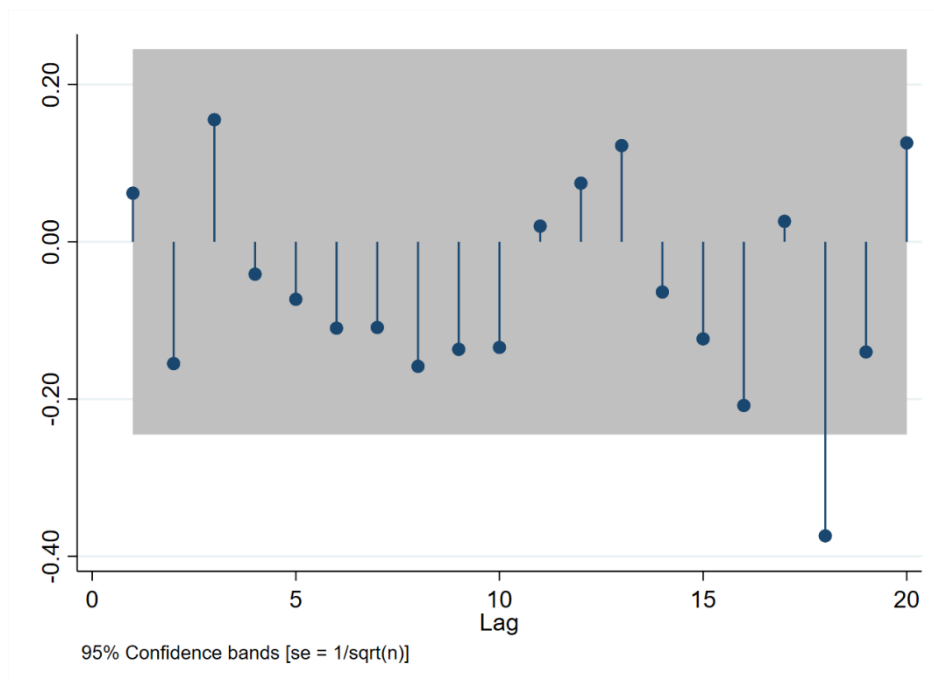


Figure 5: Partial autocorrelation diagram

Since the shape of the autocorrelation and partial autocorrelation plots is not obvious, we need to make a judgment with the help of the red pool information (AIC)

and Bayesian information (BIC), and the results are shown in Table 8. It can be found that the AIC and BIC are the smallest when P and Q are 0 and D is 2. Therefore, the autoregression of the original series with ARIMA (0,2,0) will be carried out subsequently in this paper.

Table 8: Regression information criterion of ARIMA

ID	P	D	Q	AIC	BIC
1	0	2	0	1092.89	1097.208
2	0	2	1	1094.533	1101.009
3	1	2	0	1094.649	1101.125
4	1	2	1	1094.352	1102.987

The regression results are shown in Table 9, and it can be found that the AIC and BIC values are consistent with the previous calculations, and the degree of R2 of the regression fit is 0.975, which shows that the fit is very good. The fitted and predicted values are shown in Figure 6, and it can be found that both the fitted values of the training set and the predicted values of the test set are relatively close to the true values, which are consistent with the trend of the time series.

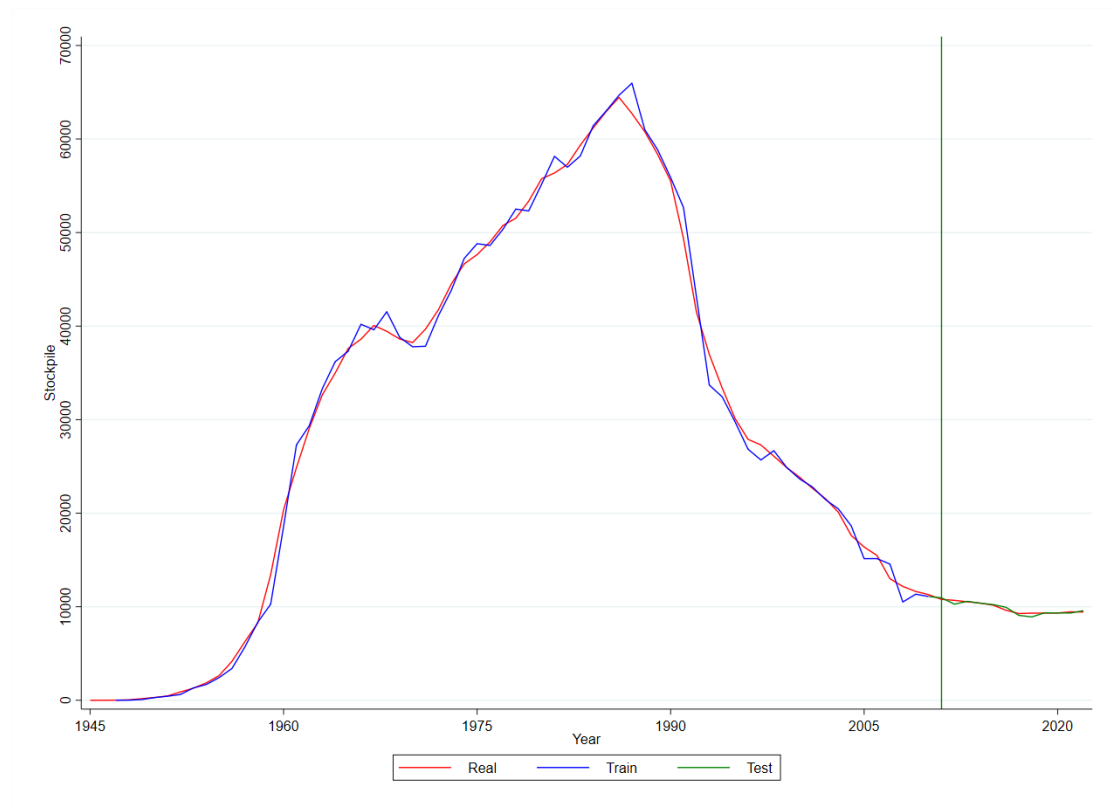


Figure 6: Prediction results of ARIMA

Table 9: Regression results of ARIMA

Item	Symbols	Value
Degree of freedom	Df Residuals	63
Number of samples	N	66
Q-statistic	Q6(P-value)	0.256(0.613)
	Q12(P-value)	4.489(0.611)
	Q18(P-value)	9.655(0.646)
	Q24(P-value)	13.883(0.737)
	Q30(P-value)	22.581(0.545)
Information Guidelines	AIC	1092.89
	BIC	1097.208
Goodness of fit	R^2	0.975

(2) LSTM model

Long Short Term Memory networks (hereinafter referred to as LSTM), a special type of RNN network, which was designed to solve the long dependency problem. It was introduced by Hoch Reiter & Schmid Huber (1997), and has been improved and popularized by many others. Their work has been used to solve a wide variety of problems and is widely used until now^[6].



Figure 7: Prediction results of LSTM

After training the training set and predicting the test set, we obtained the fitted and

predicted values and plotted them with the actual values on Fig. 7. It can be found that the fitting effect and prediction effect of LSMT is also relatively good, and its R^2 is calculated to be 0.950, which is slightly worse compared to the AIRMA model.

(3) CEEMDAN-LSTM model

Fully integrated empirical modal decomposition with Adaptive Noise (Complete E EMD with Adaptive Noise) is a new adaptive decomposition algorithm based on EMD theory and formed after improvement^[7]. Through continuous innovation, it not only overcomes the modal mixing phenomenon generated by EMD algorithm, but also improves the EMD algorithm by adding adaptive white noise, so that the reconstructed signal error is minimized. On this basis, the training prediction is then performed by LSMT model.

After the CEEMDAN decomposition to reduce the error, and then the training prediction by LSMT model, we obtained the results, as shown in Figure 8, it can be found that the fitting effect and prediction effect of CEEMDAN-LSMT is also better, and after calculation, its R^2 is 0.945, which is worse compared with LSMT model, so it can be seen that the model is not suitable for the prediction of this sequence.

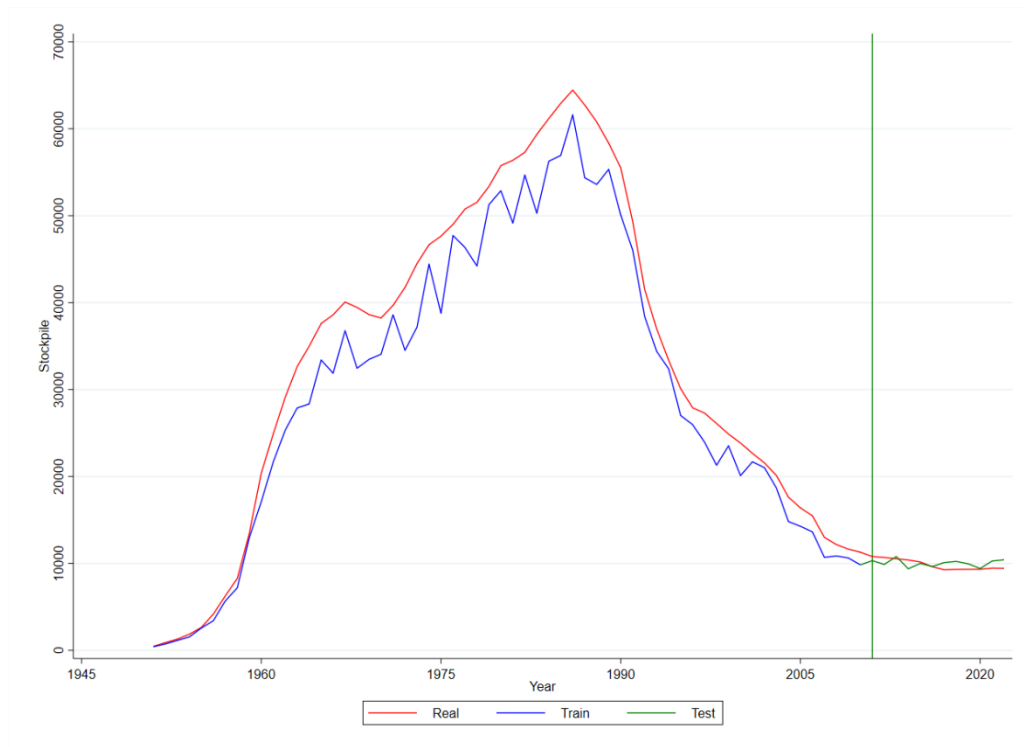


Figure 8: Prediction results of CEEMDAN-LSTM

After the above analysis, we finally decided to use ARIMA model to forecast the number of nuclear weapons, which has reached the best prediction result.

In addition, according to the above elaboration, we give the nuclear-armed countries in the next 100 years, as shown in Table 10. There are 9 of them, namely China, France, India, Israel, North Korea, Pakistan, Russia, United Kingdom and United States. The reason South Africa is excluded because it stopped possessing nuclear weapons in 1989, why according to the estimation of the world situation, it will not possess nuclear weapons in the future, so it is excluded.

Table10: The nuclear-armed nations of the next 100 years

Country	Abbreviation	First possession of nuclear weapons time
China	CHN	1964
France	FRA	1960
India	IND	1987
Israel	ISR	1967
North Korea	PRK	2006
Pakistan	PAK	1987
Russia	RUS	1949
United Kingdom	GBR	1952
United States	USA	1945

4.2.2 Sub-question 2

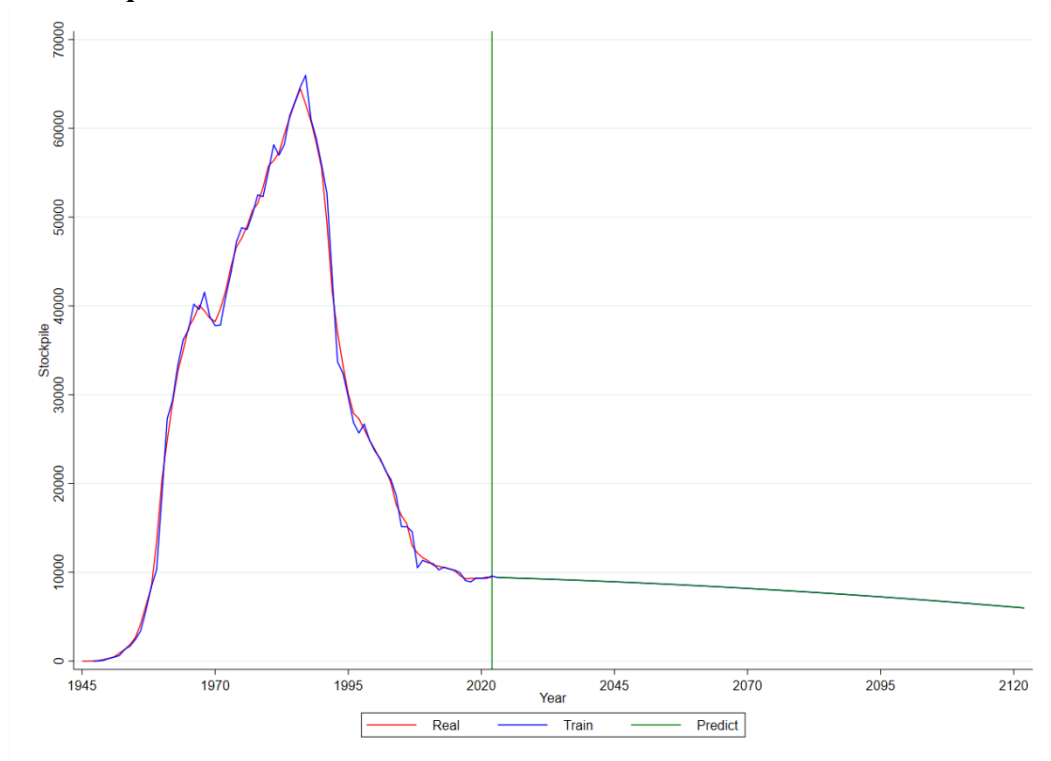


Figure 9: Total number of nuclear weapons in the next 100 years

Based on the model given in the previous sub-question, we predicted the global trend of the number of nuclear weapons in the next 100 years by ARIMA model, and the results are shown in Figure 9. It can be found that the fitting effect is good, and the overall trend is decreasing, and after reaching a peak in the 1990s, it keeps decreasing and will continue this trend after 2022 as well.

In addition, we also predict the number of nuclear weapons of the nine nuclear-armed countries in the world after 101 years, i.e., 2123, by ARIMA model, and the results are shown in Figure 10 and Table 11. It can be found that the number of nuclear weapons changes of most nuclear-armed countries follow the global general trend, while some nuclear-armed countries keep the same number of nuclear weapons, such as Pakistan, while the United States and the United Kingdom even destroy all their nuclear weapons, and finally the total number of nuclear weapons of the nine nuclear-armed countries reaches 5518 in 2123.

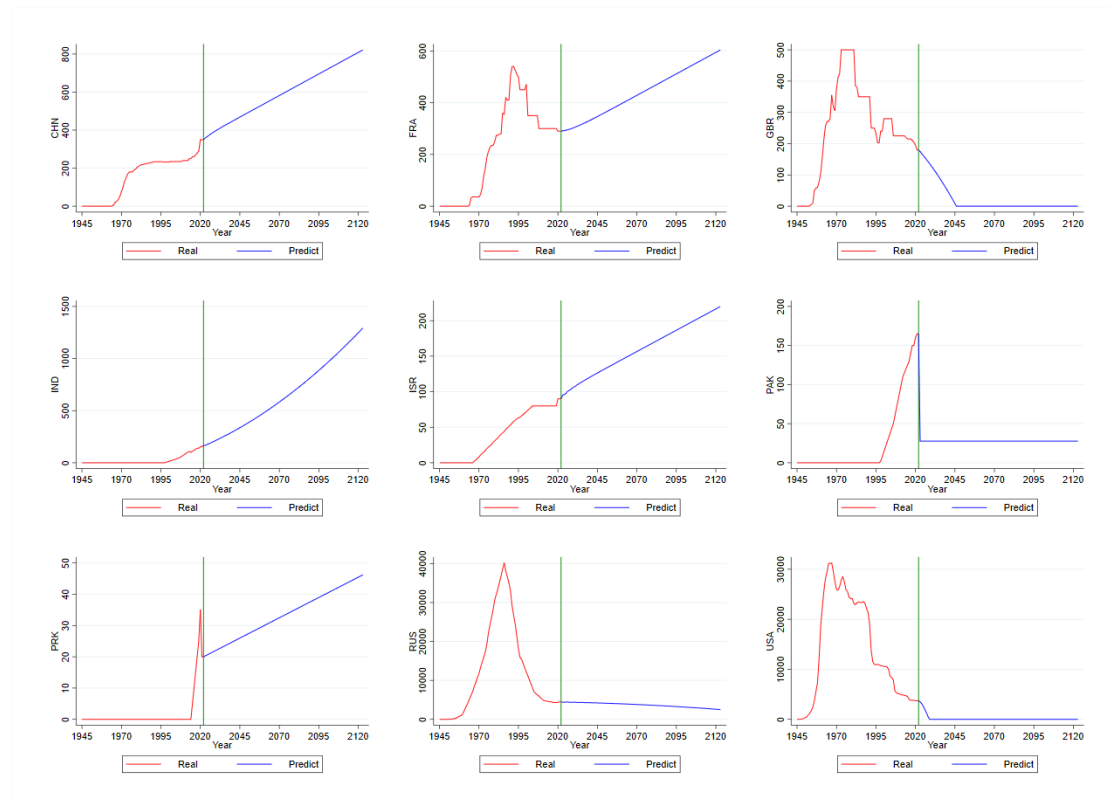
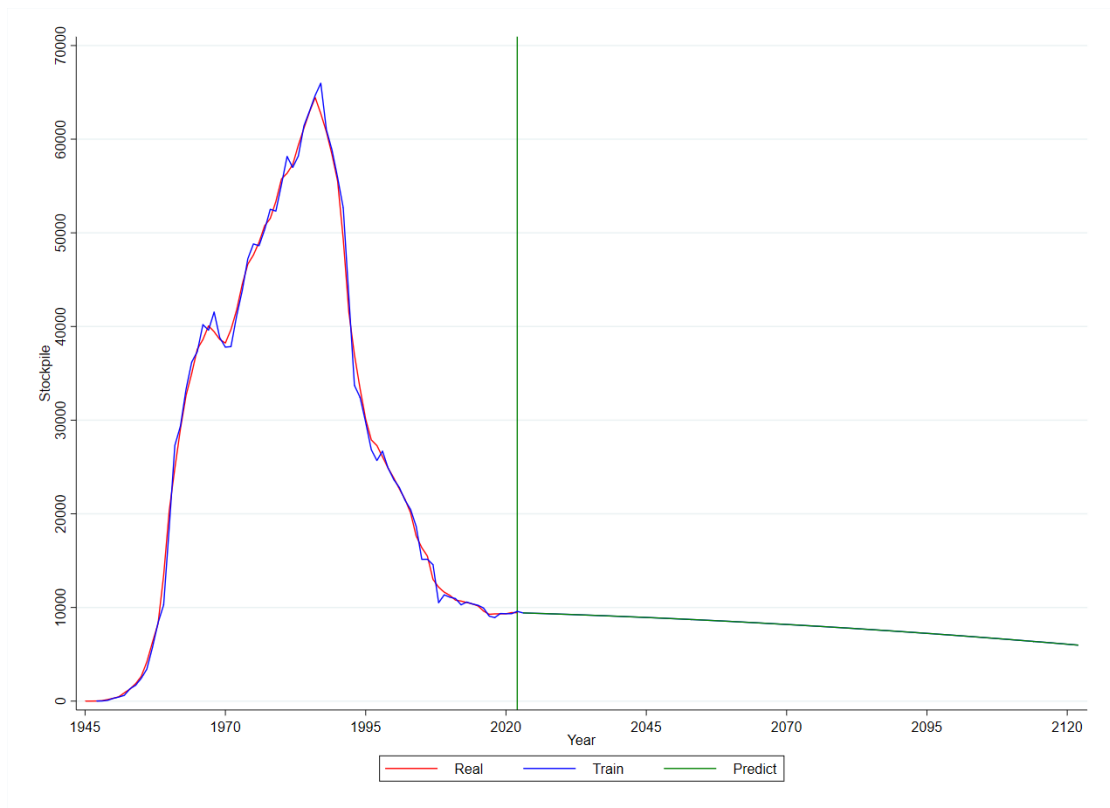


Figure 10: The number of nuclear weapons of each country in the next 101 years

Table 11: The number of nuclear weapons in each country in 2123

Country	Abbreviation	stockpiles
United States	USA	0
United Kingdom	GBR	0
Pakistan	PAK	28
North Korea	PRK	46
Israel	ISR	220
France	FRA	604
China	CHN	822
India	IND	1293
Russia	RUS	2504
Total		5518

In order to check the prediction effect of the model, we use the global number of nuclear weapons to predict the total number of nuclear weapons in the world in 2123, the result is shown in Figure 11, the specific number is 5927, the above prediction result is not significantly different, so we consider the above prediction result is credible, and use it as our predicted number of nuclear weapons for each nuclear-armed country in 2123, the total number of nuclear weapons in the world is 5518.

*Figure 11: The number of nuclear weapons globally in the next 101 years*

4.3 Solution of Questions 3

Regarding this question, we define the destruction of 1% of the human habitable land area as the destruction of the human living environment and 5% of the human habitable land area as the complete destruction of the Earth, based on the assumptions in the previous section.

4.3.1 Sub-question 1

First, we consulted the data and collected the land area of each continent as shown in Table 12. Then, we gave the proportion of habitable land area for each continent based on the literature study as shown in the table, and then we calculated the habitable land area of each continent and the whole Earth. The areas of each continent are shown in Figure 12.

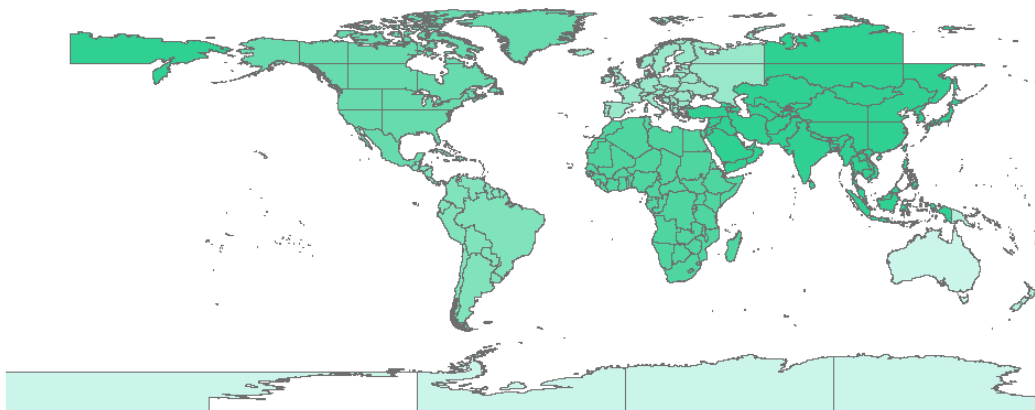


Figure 12: Distribution by continent according to area

Secondly, according to the information we have inquired, the largest nuclear bomb on earth is the "Tsar Bomb", its TNT equivalent is 57 million tons, the explosion destruction radius is 18 kilometers, so we calculate that its destruction area is about 0.1018 million square kilometers, and the total habitable land area of the earth is 106.7 million square kilometers. Assuming that the power of each nuclear bomb is comparable to that of the "Tsar Bomb", at least 5241 nuclear bombs will be needed to completely

destroy the human environment.

Table 12: Estimates of habitable area by continent

Continental	Area	Livable Proportion	Livable Area
Asia	4400	80%	3520
Europe	1000	75%	750
North America	2400	85%	2040
South America	1800	70%	1260
Africa	3000	70%	2100
Oceania	900	80%	720
Antarctica	1400	20%	280
Total	14900	-	10670

4.3.2 Sub-question 2

Following the previous question, there are 9440 nuclear bombs on Earth in 2022, and the same assumption is that the power of nuclear bombs is comparable to that of the "Tsar Bomb", then a total of 9.609 million square kilometers of habitable land can be destroyed, as shown in Table 13. This is enough to destroy the living environment of human beings, which is means to destroy the Earth.

Table 13: Current Nuclear Weapons Killing Power on Earth

Continental	Livable Area	Area destroyed	Ratio
Asia	3520	316.989673	9.01%
Europe	750	67.54041328	9.01%
North America	2040	183.7099241	9.01%
South America	1260	113.4678943	9.01%
Africa	2100	189.1131572	9.01%
Oceania	720	64.83879675	9.01%
Antarctica	280	25.21508762	9.01%
Total	10670	960.8749463	9.01%

4.3.3 Sub-question 3

In order to protect the living environment of human beings, we can only allow nuclear weapons to destroy less than 1% of the total land area, which means we can only endanger 1.067 million square kilometers of habitable land area, which is means we can only have a maximum of 1048 nuclear bombs with the same power as the "Tsar Bomb", so we give and nuclear weapons limit The total number of nuclear weapons is 1048, which is divided into each nuclear-armed country according to the proportion of

nuclear bombs held by nuclear-armed countries in 2022, as shown in Table 14.

Table 14: Limitations on the number of nuclear weapons in each nuclear-armed state

Country	Available Quantity	Limit the number of	Ratio
China	350	39	11%
France	290	32	11%
India	160	18	11%
Israel	90	10	11%
North Korea	20	2	11%
Pakistan	165	18	11%
Russia	4477	492	11%
United Kingdom	180	20	11%
United States	3708	408	11%
Total	9440	1038	11%

4.4 Solution of Questions 4

Since the first mushroom cloud of successful nuclear weapons tests appeared, nuclear war has become the greatest threat to human . The immense destructive power of nuclear weapons has led to a worldwide consensus that a world with nuclear weapons is very dangerous. Humankind must make a concerted effort to accomplish the total elimination of nuclear weapons.

However, as a result of the international confrontation characterized by the Cold War after the end of World War II, an endless nuclear arms race led to the creation of a vast nuclear arsenal. This nuclear arsenal is a great threat to both nuclear-weapon and non-nuclear-weapon States, because any nuclear war or even nuclear accident is unbearable for mankind. This situation has led mankind to ponder a serious question: how to fundamentally eliminate the threat posed by nuclear weapons to mankind?

So this paper draws the following conclusions by analyzing the data of a batch of nuclear weapons.

Firstly, the overall trend of the world tends to be peaceful, and the number of nuclear weapons on earth has reached a growth peak, is currently declining, and will continue this trend in the future.

Secondly, in the future, there are only 9 nuclear-armed countries, in addition to the permanent members of the United Nations Security Council, there are also 4 countries, India, Israel, North Korea, and Pakistan, and according to the prediction, the number of nuclear weapons in most of these 9 countries will tend to decline in the future, even to 0, and completely release the possession of nuclear weapons, which has an important positive effect on the survival and development of human beings in the future.

Finally, although the current trend is in a good direction and the future estimates are optimistic, the number of nuclear weapons currently possessed by human beings is still relatively high, and if all of them are used to strike cities where human beings gather with precision, then the survival environment of human beings will suffer unprecedented harm, and each nuclear-armed country needs to further reduce its own nuclear weapons stockpile.

We make the following recommendations for the nuclear-armed states.

Firstly, countries join together to sign nuclear ban convention

Thirdly, countries do not use nuclear weapons unless necessary.

Thirdly, countries do not be the first to provoke war or use nuclear weapons.

Fourthly, countries do not use nuclear weapons against non-nuclear countries.

Fifthly, countries not to provide nuclear weapons technology to non-nuclear states.

Sixthly, countries reduce nuclear weapons stockpiles to 11 percent of current levels as much as possible while maintaining national security.

References

- [1] ARONR, MAHONEY D J, ANDERSON B C. Peace & War: A theory of international relations [M] . London; New York: outledge2017: 407
- [2] LIU Tetra Le. Journal of the University of South China(Social Science Edition),2022,23(04):1-8.DOI:10.13967/j.cnki.nhxb.2022.0046.)
- [3] YANG Xiuyuan, XIAO Yang, CHEN Shuyong. Prediction of wind speed and power generation in wind farm[J].Proceedings of the CSEE,2005(11):1-5.)
- [4] TranDT, IosifidisA, Kannianen [J], etal. Temporal Attention-Augmented Bilinear Network for Financial Time-Series Data Analysis[J]. IEEE Transactionson Neural Networksand Learning Systems,2019,30(99).
- [5] CUI Jianguo, LI Pengcheng, CUI Xiao, YU Mingyue, JIANG Liying, WANG Jinglin. Performance trend prediction method of aircraft hydraulic pump based on ARIMA-LSTM[J].Vibration. Testing and Diagnostics,2021,41(04):735-740+832.DOI:10.16450/j.cnki.issn.1004-6801.2021.04.015.
- [6] FANG Qian, YANG Nan. Construction of nonlinear error correction learning model based on LSTM network[J/OL].Statistics and Decision,2022(24):5-10[2023-01-08]. DOI:10.13546/j.cnki.tjyjc.2022.24.001.
- [7] SHEN Fuxin,BING Qichun,ZHANG Weijian,HU Yanran,GAO Peng. Short-term traffic flow prediction based on CEEMDAN-ABC-LSTM combination model[J].Journal of Qingdao University of Technology,2022,43(05):96-103+119.)

Appendix

(1) Python 代码

```
import os
import random
import numpy as np
import pandas as pd
import tensorflow as tf
from copy import deepcopy
from keras import backend
from PyEMD.CEEMDAN import CEEMDAN
from matplotlib import pyplot as plt
from tsfresh import extract_features
from sklearn.linear_model import Lasso
from sklearn.preprocessing import StandardScaler
from tsfresh.utilities.dataframe_functions import impute
from tsfresh.utilities.dataframe_functions import roll_time_series

# 设置全局随机性, 尽可能实现结果复现
os.environ["CUDA_VISIBLE_DEVICES"] = "-1"
seed_value = 1234567
os.environ['PYTHONHASHSEED'] = str(seed_value)
random.seed(seed_value)
np.random.seed(seed_value)
tf.compat.v1.set_random_seed(seed_value)
session_conf =
tf.compat.v1.ConfigProto(intra_op_parallelism_threads=1,
inter_op_parallelism_threads=1)
sess = tf.compat.v1.Session(graph=tf.compat.v1.get_default_graph(),
config=session_conf)
backend.set_session(sess)

def ceemdan(signal, time):
    # 设置试验次数
    ceemdan = CEEMDAN(trials=100)

    # 最低筛选次数
    ceemdan.FIXE_H = 100

    eIMFs = ceemdan(signal, time, -1)

    imfs, res = ceemdan.get_imfs_and_residue()
```

```
nIMFs = imfs.shape[0]

# 可视化
plt.figure(figsize=(12, 9))
plt.subplot(nIMFs + 1, 1, 1)
plt.plot(time, signal, 'r')

for n in range(nIMFs):
    plt.subplot(nIMFs + 1, 1, n + 2)
    plt.plot(time, eIMFs[n], 'g')
    plt.ylabel("eIMF %i" % (n + 1))
    plt.locator_params(axis='y', nbins=5)

plt.xlabel("Time")
plt.tight_layout()
plt.savefig('ceemdan_example', dpi=120)
plt.show()

summm = sum(eIMFs[1:])

plt.plot(summm, color='green', label='eemd 合并')
plt.title('eemd 合并')
plt.xlabel('时间')
plt.ylabel('金融数据')
plt.legend()
plt.show()plt.show()

eIMFs_pd = pd.DataFrame(eIMFs.T)

return eIMFs_pd

def data_exchange(data_raw, time):
    data_exchanged = data_raw
    data_exchanged['Symbols'] = np.array('element').repeat(108).T
    data_exchanged['id'] = np.array(np.nan).repeat(108).T
    data_exchanged['date'] = time

    return data_exchanged

def X_tsfresh(ts_data):
    # 双倍数据集
    data_double = pd.concat([ts_data, ts_data])
```

```
# 滚动得到原始数据集转换集
df_rolled = roll_time_series(data_double, column_id="Symbols",
column_sort="date",
                                max_timeshift=1, min_timeshift=1,
rolling_direction=2)

del df_rolled['Symbols']

# 提取特征
X = extract_features(df_rolled,
                    column_id="id",
                    column_sort="date",
                    impute_function=impute,
                    show_warnings=False)

X = X.set_index(X.index.map(lambda x: x[1]), drop=True)

return X

# 去除 x_ 数据集重复维度
def X_Uniqueness(x_vls):
    # 先转置去重
    x_vls_T = x_vls.T
    x_vls_new = np.array([x_vls_T[0]])

    for i in range(len(x_vls_T)):
        ii = x_vls_T[i]
        if not (x_vls_new == ii).all(1).any():
            x_vls_new = np.row_stack((x_vls_new, ii))

    # 再转回来
    x_vls_new = x_vls_new.T

    return x_vls_new

def X_lasso(signal, x_vls_new, time):
    # 建立滞后
    x_data = x_vls_new[:-18]
    y_data = signal[18:]

    # 标准化
    scaler = StandardScaler()
```

```
x_data_scaler = scaler.fit_transform(x_data)

# 声明模型, fit!!!!!!
model = Lasso()
model.fit(x_data_scaler, y_data)

# 预测, 可以不用预测, 保险起见还是看一眼
y_pre = model.predict(x_data_scaler)
plt.plot(y_pre, color='red')
plt.plot(y_data, color='green')
plt.title('train')
plt.legend()
plt.show()

# 得到 lasso 筛选结果, 提取非 0 权重向量
coef = model.coef_
x_vls_new = x_vls_new.T
x_lasso_ay = np.array([x_vls_new[0]])

for i in range(len(coef)):
    ii = x_vls_new[i]
    if coef[i] != 0:
        x_lasso_ay = np.row_stack((x_lasso_ay, ii))

# 生成 dataframe, 添加时间索引
x_lasso_pd = pd.DataFrame(x_lasso_ay[1:].T)
x_lasso_pd['date'] = time
x_lasso_pd = x_lasso_pd.set_index('date')
return x_lasso_pd

# x 数据集处理集成
def X_Collect(signal, time):
    # ceemdan 将原始数据集在不同频段上分解, 得到多个周期和趋势
    X_ced = (ceemdan(signal=signal, time=time))
    X_ced_ay = np.array(deepcopy(X_ced).values)

    # 数据转换, 以符合 tsfresh 数据处理的格式
    X_exed = data_exchange(X_ced, time)

    # tsfresh 暴力生成大量特征
    X_tsed = X_tsfresh(X_exed)
    X_tsed_ay = np.array(X_tsed.values)
```

```
# 合并数据集: ceemdan+tsfresh
x_vls = np.concatenate((X_ced_ay, X_tsed_ay), axis=1)

# 重复特征剔除
x_vls_new = X_Uniqueness(x_vls=x_vls)

# 利用 lasso 选择特征
x_data_lasso = X_lasso(signal=signal, x_vls_new=x_vls_new,
time=time)

return x_data_lasso

def Y_Collect(signal, time):
    # 以 18 个月为窗口滚动, 形成 y 的多输出数据集
    Y_exed_ay = np.array([np.arange(0, 18)])
    for i in range(len(signal)):
        Y_exed_ay_i = signal[i:i + 18]
        if len(Y_exed_ay_i) == 18:
            Y_exed_ay = np.row_stack((Y_exed_ay, Y_exed_ay_i))

    # 提取最后 90 个窗口, 剔除不足 18 个维度的样本, 并对齐 x 数据集
    Y_exed = Y_exed_ay[-90:]

    # 增加空集, 补齐数据长度, 添加时间索引
    for i in range(18):
        Y_exed = np.row_stack((Y_exed, np.array(np.nan).repeat(18)))

    Y_pd = pd.DataFrame(Y_exed)
    Y_pd['date'] = time

    Y_pd = Y_pd.set_index('date')
    return Y_pd

if __name__ == "__main__":
    data = pd.read_excel('./data/e_raw_data.xlsx')

    # 初始数据提取
    signal = data['e_x_raw'].values
    time = data['date'].values

    # x 数据集生成
    X_data = X_Collect(signal=signal, time=time)
```

```
X_data.to_excel('./data/Backup/X_data.xlsx')

# Y 数据集生成
Y_data = Y_Collect(signal=signal, time=time)
Y_data.to_excel('./data/Backup/Y_data.xlsx')

print(X_data)
print(Y_data)
```

(2) Stata 代码

```
cd D:\ZM\亚太杯\E

import excel 2022_APMCM_E_Data.xlsx,sheet(proliferation) firstrow clear
save odata1.dta,replace

import excel 2022_APMCM_E_Data.xlsx,sheet(stockpiles) firstrow clear
save odata2.dta,replace

import excel 2022_APMCM_E_Data.xlsx,sheet(tests) firstrow clear
save odata3.dta,replace

import excel 2022_APMCM_E_Data.xlsx,sheet(position) firstrow clear
save odata4.dta,replace

use odata1.dta,clear

use odata2.dta,clear

use odata3.dta,clear

use odata4.dta,clear

//第 1 题第 1 问
use odata4.dta,clear
```

```

keep if Status == 3
bys Country:gen n = _n
keep if n == 1

//第 1 题第 2 问
use odata2.dta,clear
keep if Year == 2002 | Year == 2022
sort Country Year
gen lack = Stockpile - Stockpile[_n-1] if Year == 2022
keep if Year == 2022
sort lack

graph hbar (sum) lack,over(Country,sort(1) label(labsize(*0.5))) ytitle("Number
of changes") graphregion(color(white))
graph export D:\ZM\亚太杯\图\F1-2.tif,replace

//第 1 题第 3 问
use odata3.dta,clear
bys Year:egen test = sum(Tests)
bys Year:gen n = _n
keep if n == 1
drop n
gen t_add5 = test + test[_n+1] + test[_n+2] + test[_n+3] + test[_n+4]
sort t_add5
sort Year

line test t_add5 Year if Year<2016,xline(1962,lcolor(green))ylabel(0) xtitle("Year")
ytitle("Test Time") legend(label(1 "One year") label(2 "Five year"))
graphregion(color(white)) scale(0.8)
graph export D:\ZM\亚太杯\图\F1-3.tif,replace

```

//第 1 题第 4 问

```
use odata3.dta,clear
```

```
keep if Year > 2011
```

```
bys Country: egen Test_add10 = sum(Tests)
```

```
bys Country: gen n = _n
```

```
keep if n == 1
```

```
drop n
```

```
sort Test_add10
```

//第 1 题第 5 问

```
use odata4.dta,clear
```

```
keep if Status == 0 | Status == 3
```

```
bys Country:egen t1 = max(Year) if Status == 0
```

```
bys Country:egen t2 = min(Year) if Status == 3
```

```
bys Country:egen s1 = max(t1)
```

```
bys Country:egen s2 = min(t2)
```

```
keep if s2 != .
```

```
gen lack = s2-s1
```

```
bys Country:gen n = _n
```

```
keep if n == 1
```

```
sort lack
```

```
drop if Abbreviation == "ZAF"
```

```
graph hbar (sum) lack,over(Country,sort(1) label(labsize(*0.5))) ytitle("Year  
spent") graphregion(color(white))
```

```
graph export D:\ZM\亚太杯\图\F1-5.tif,replace
```

//第 2 题第 1 问

```
use odata2.dta,clear
```

```
bys Year: egen all_stock = sum(Stockpile)
```

```
bys Year: gen n = _n
```



```
keep if n == 1
drop n
keep Year Stockpile all_stock
save data1.dta,replace

//ARIMA 预测
import excel 每年核弹数量.xlsx,firstrow clear
tsset Year
keep if Year < 2011
dfuller all_stock
gen all_stock_1 = d.all_stock
dfuller all_stock_1
gen all_stock_2 = d2.all_stock
dfuller all_stock_2

ac all_stock_2, lags(20) level(95) ytitle("") graphregion(color(white))
graph export D:\ZM\亚太杯\图\F2-1-1-1.tif,replace
pac all_stock_2, lags(20) level(95) ytitle("") graphregion(color(white))
graph export D:\ZM\亚太杯\图\F2-1-1-2.tif,replace

arima all_stock,arima(0,2,0)
estat ic
arima all_stock,arima(0,2,1)
estat ic
arima all_stock,arima(1,2,0)
estat ic
arima all_stock,arima(1,2,1)
estat ic
```

```
import excel 每年核弹数量.xlsx,firstrow clear
tsset Year
arima all_stock if Year < 2011 ,arima(0,2,0)
predict y_hat,y
```

```
twoway (line all_stock Year if Year <2023,color(red)) (line y_hat Year if
Year<2011,color(blue)) (line y_hat Year if Year>2009 & Year <
2023,color(green)),xline(2011,lcolor(green)) xlabel(1945(15)2022)
ylabel(0(10000)70000) xtitle("Year") ytitle("Stockpile") legend(label(1 "Real") label(2
"Train") label(3 "Test") col(3)) graphregion(color(white)) scale(0.5)

graph export D:\ZM\亚太杯\图\F2-1-1-3.tif,replace
```

```
//LSTM 预测
```

```
import excel 预测 LSTM+EEMDLSTM.xlsx,firstrow clear
replace D = _n + 1950
rename D Year
rename y all_stock
rename lstm y_hat
```

```
twoway (line all_stock Year if Year <2023,color(red)) (line y_hat Year if
Year<2011,color(blue)) (line y_hat Year if Year>2009 & Year <
2023,color(green)),xline(2011,lcolor(green)) xlabel(1945(15)2022)
ylabel(0(10000)70000) xtitle("Year") ytitle("Stockpile") legend(label(1 "Real") label(2
"Train") label(3 "Test") col(3)) graphregion(color(white)) scale(0.5)

graph export D:\ZM\亚太杯\图\F2-1-2.tif,replace
```

```
//CEEMDAN-LSTM 预测
```

```
import excel 预测 LSTM+EEMDLSTM.xlsx,firstrow clear
replace D = _n + 1950
rename D Year
rename y all_stock
```

```

rename eemd y_hat

twoway (line all_stock Year if Year <2023,color(red)) (line y_hat Year if
Year<2011,color(blue)) (line y_hat Year if Year>2009 & Year <
2023,color(green)),xline(2011,lcolor(green)) xlabel(1945(15)2022)
ylabel(0(10000)70000) xtitle("Year") ytitle("Stockpile") legend(label(1 "Real") label(2
"Train") label(3 "Test") col(3)) graphregion(color(white)) scale(0.5)

graph export D:\ZM\亚太杯\图\F2-1-3.tif,replace

```

```

//第 2 题第 2 问

import excel 每年核弹数量.xlsx,firstrow clear

tsset Year

arima all_stock if Year < 2011 ,arima(0,2,0)

predict y_hat,y

twoway (line all_stock Year if Year <2023,color(red)) (line y_hat Year,color(blue))
(line all_stock Year if Year > 2022,color(green)),xline(2022,lcolor(green))
xlabel(1945(25)2122) ylabel(0(10000)70000) xtitle("Year") ytitle("Stockpile")
legend(label(1 "Real") label(2 "Train") label(3 "Predict") col(3))
graphregion(color(white)) scale(0.5)

graph export D:\ZM\亚太杯\图\F2-2-1.tif,replace

```

```

//第 2 题第 2 问各个国家

use odata2.dta,clear

bys Year Country: egen all_stock = sum(Stockpile)

sort Country Year

tab Abbreviation

global xlist "CHN FRA GBR IND ISR PAK PRK RUS USA"

foreach x of global xlist{
    export excel Year all_stock if Abbreviation == "`x'" using
`x'_p.xlsx,firstrow(variables) replace
}

```

```

export excel Year all_stock if Abbreviation == "ZAF" using
ZAF_p.xlsx,firstrow(variables) replace

```

```

import excel ZAF_p.xlsx,firstrow clear
gen Country = "ZAF"
save data2_2.dta,replace
foreach x of global xlist{
import excel `x'_p.xlsx,firstrow clear
gen Country = "`x'"
append using data2_2.dta
save data2_2.dta,replace
}
use data2_2.dta,clear
rename all_stock stockpiles
replace stockpiles = 0 if stockpiles < 0
br if Year == 2123
//bys Year:egen all_stock = sum(stockpiles)
//bys Year:gen n = _n
//keep if n == 1

foreach x of global xlist{
twoway (line stockpiles Year if Year < 2023 & Country == "`x'",color(red)) (line
stockpiles Year if Year >= 2022 & Country ==
"`x'",color(blue)),xline(2022,lcolor(green)) xlabel(1945(25)2122) ylabel()
xtitle("Year") ytitle("`x'") legend(label(1 "Real") label(2 "Predict") col(2))
graphregion(color(white)) scale(0.5)
graph save `x'.gph,replace
}
graph combine CHN.gph FRA.gph GBR.gph IND.gph ISR.gph PAK.gph
PRK.gph RUS.gph USA.gph,graphregion(color(white))

```

```
graph export D:\ZM\亚太杯\图\F2-2-2.tif,replace
```

```
import excel 每年核弹数量.xlsx,firstrow clear
tsset Year
arma all_stock if Year < 2023 ,arma(0,2,0)
predict y_hat,y
twoway (line all_stock Year if Year < 2023,color(red)) (line y_hat Year,color(blue))
(line all_stock Year if Year > 2022,color(green)),xline(2022,lcolor(green))
xlabel(1945(25)2122) ylabel(0(10000)70000) xtitle("Year") ytitle("Stockpile")
legend(label(1 "Real") label(2 "Train") label(3 "Predict") col(3))
graphregion(color(white)) scale(0.5)

graph export D:\ZM\亚太杯\图\F2-2-3.tif,replace
```

```
//第3题第1问
```

```
use odata2.dta,clear
bys Year: egen all_stock = sum(Stockpile)
bys Year: gen n = _n
keep if n == 1
drop n
keep Year all_stock
br if Year == 2022
```

```
import excel GTA.xlsx,firstrow clear
drop in 1/2
bys CuntryCode:gen n = _n
keep if n == 1
destring CuntryCode,replace
sort CuntryCode
drop in 170/174
drop in 1
```